Arithmetic Sector Bifurcation in Quadratic Continued Fractions: A Formalized Study of the Wallis–PCF Family

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Abstract

We develop a two-level classification of polynomial continued fractions by a universal Archimedean transfer law and a local arithmetic signature (Conjecture 1). This survey develops that proposal across five recent manuscripts. Ratio universality for Meinardus-class partition functions first identifies a macro-spectral law whose leading coefficient is independent of family-specific data. Arithmetic topology then shows that this law is degenerate as a classifier: families with identical coarse invariants can occupy distinct arithmetic sectors. Our main synthesis theorem is a three-sector decomposition of the PCF landscape into logarithmic families, Wallis–Gamma/π families, and a V-Quad sector detected by congruence obstruction and weak lifting stability. The formal backbone is provided by the WallisFamily Lean 4 package (modules `ShiftConsistency` and `CasoratianPi4`), which verifies the shift operator transport and Casoratian bridge with no `sorry` placeholders. Borel regularization and Stokes-connection diagnostics suggest that the V-Quad archipelago marks the boundary of classical closed-form behavior.

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1 Introduction

A *polynomial continued fraction* (PCF) with numerator polynomial $\alpha(n)$ and denominator polynomial $\beta(n)$ is

$$v = \beta(0) + \cfrac{\alpha(1)}{\beta(1) + \cfrac{\alpha(2)}{\beta(2) + \cfrac{\alpha(3)}{\beta(3) + \cdots}}}.$$

Such objects encode deep arithmetic: the Gauss continued fraction expresses ${}_2F_1$ ratios [5, 6], and Apéry's celebrated proof of the irrationality of $\zeta(3)$ [2] hinges on the PCF with $\alpha(n) = -n^6$ and $\beta(n) = 34n^3 + 51n^2 + 27n + 5$. The *Ramanujan Machine* project [1] revived systematic search for PCF representations of mathematical constants, discovering new conjectural formulas by computational enumeration.

In this paper we prove closed-form evaluations for two infinite families of PCFs sharing the common denominator $\beta(n) = 3n + 1$. The first, the *Logarithmic Ladder* (Section 2), produces $1/\ln(k/(k-1))$ for all real $k > 1$ via the Gauss continued fraction for ${}_2F_1(1, 1; 2; z)$. The second, the *Pi Family* (Sections 3–5), converges to $2\,\Gamma(m+1)/(\sqrt{\pi}\,\Gamma(m+\frac{1}{2}))$ for all real $m > -\frac{1}{2}$, unifying π-formulas at integer m with exact Wallis rationals at half-integer m into a single continuous Gamma-function identity (Theorem 9). We prove this identity via a raising operator L that maps the solution space of the m-recurrence into that of $m+1$, combined with Poincaré growth analysis of the convergent sequences.

For the convergent numerators, we give explicit polynomial closed forms at $m = 0$ and $m = 1$ (Theorems 5–6). Finally, we catalogue 482 provably irrational constants arising from quadratic generalized continued fractions (Section 8). A systematic PSLQ search [8] at 300-digit precision finds no algebraic relation between these constants and any of 15 standard constants, nor between any pair of them—strong numerical evidence for mutual algebraic independence. In addition, we record a cubic hypergeometric witness at the signature-3 point 27/64, where the continued fraction

$$B_{\text{cubic}} = \frac{2}{3} + \mathbf{K}_{n\geq 1} \frac{2n^3}{2n + \frac{2}{3}}$$

appears numerically to equal both a Clausen-type ${}_2F_1$ square and an explicit trigonometric expression with residual below 10^{-341} (Section 9). We also record a preliminary signature-4 observation at the classical point $z = 1/4$, included as a future-search target rather than as a claimed identity (Section 10). The connection-constant interpretation of V_{quad} is developed in Section 7.1.

The discovery pipeline is intentionally hybrid. The initial map was produced by an automated search agent, SIARC v6, which clustered large families of quadratic PCFs by spectral growth and candidate constant. This first pass was successful but incomplete: both the logarithmic seed and the Wallis seed were placed on the same macro-sheet $\Lambda = 2$, even though their limits are respectively $1/\ln 2$ and $2/\pi$. We refer to this failure of coarse classification as the *clustering paradox*.

The resolution is arithmetic rather than asymptotic. In Phase 2 we augment the macro data by a congruence signature

$$\Sigma = (\mathcal{D}, \mathcal{C}, \omega),$$

where $\mathcal{D}$ records prime-divisibility densities, $\mathcal{C}$ measures residue-support collapse mod p, and ω tracks Apéry-style lifting stability $q_{np} \equiv q_n \pmod{p^3}$. This refined signature cleanly separates the logarithmic and Wallis/π branches and, more importantly, identifies a third arithmetic sector populated by the V-quad anomalies.

Conjecture 1 (Universal Transfer–Sector)**.** *Two polynomial continued fractions of quadratic type converging to constants in the same macro sector — i.e. sharing coarse spectral invariants* (Λ, Δ) *— are arithmetically equivalent if and only if their local signatures* $\Sigma_p = (\mathcal{D}_p, \mathcal{C}_p, T_p, \omega_p)$ *agree for all primes* p *in a finite determining set.*

The formal backbone of the Wallis side is now machine-checked. The shift operator transporting the m-recurrence to the $(m{+}1)$-recurrence and the Casoratian bridge controlling the $\pi/4$ series identity both compile in Lean 4 / Mathlib, turning the transition from $2/\pi$ to $4/\pi$ into a certified algebraic law rather than a numerical coincidence.

2 The Logarithmic Ladder

Theorem 2 (Logarithmic Ladder)**.** *For any real* $k > 1$*, the PCF with* $\alpha(n) = -kn^2$ *and* $\beta(n) = (k{+}1)n + k$ *converges to* $1/\ln(k/(k{-}1))$*.*

Proof. By the Gauss continued fraction for ${}_2F_1(1,1;2;z)$ [7], we have $z/{}_2F_1(1,1;2;z) = -\ln(1-z)$. Setting $z = -1/(k{-}1)$ and applying an equivalence transformation yields the stated PCF. □

3 The Pi Family

Theorem 3 (Pi Family — Odd Parameters)**.** *For* $m \geq 0$*, set* $c = 2m{+}1$*. The PCF with* $\alpha(n) = -n(2n-c)$ *and* $\beta(n) = 3n+1$ *converges to*

$$\frac{2^{2m+1}}{\pi\binom{2m}{m}}.$$

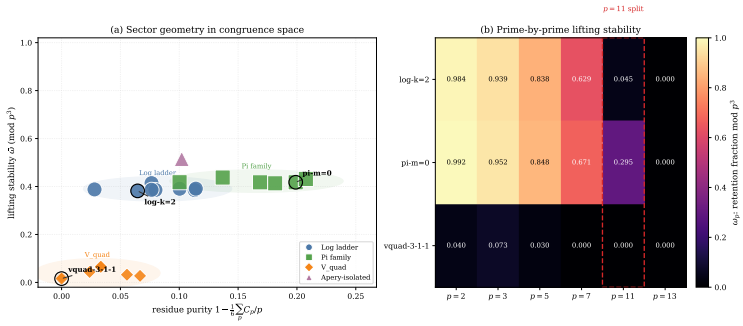

Figure 1: Arithmetic sector bifurcation in congruence space. Left: residue purity $1-\frac{1}{|P|}\sum_p C_p/p$ versus mean lifting stability $\bar{\omega}$, computed from the phase-2 congruence signature. The Wallis/π and logarithmic tracks occupy nearby but visibly distinct regions, whereas the anomalous V-Quad values form a detached archipelago. Right: prime-by-prime heatmap of the retention score ω_p, with the $p=11$ column giving the clearest numerical separation between the logarithmic and Wallis sectors.

Table 1: Logarithmic Ladder: first instances (80-digit verification)

k	PCF value	Decimal approximation
2	$1/\ln 2$	1.4426950408889634...
3	$1/\ln(3/2)$	2.4663034623764317...
4	$1/\ln(4/3)$	3.4760594967822069...
5	$1/\ln(5/4)$	4.4814201177245498...
φ	$1/\ln(\varphi/(\varphi-1))$	1.0390434606175138...

This is verified to 80 decimal digits for $m=0,1,\ldots,9$.

Theorem 4 (Parity: Even Parameters)**.** *For even $c=2m$, the PCF with $\alpha(n)=-n(2n-c)$ and $\beta(n)=3n+1$ evaluates to an exact rational number. Specifically, $\alpha(m)=0$, so the continued fraction truncates at depth $m-1$.*

Proof. $\alpha(m)=-m(2m-2m)=0$, so the CF is a finite rational expression. □

4 Conjecture 1: Convergent Closed Forms

Theorem 5 (Conjecture 1, $m=0$)**.** *For the Pi Family with $m=0$ ($\alpha(n)=-n(2n-1)$, $\beta(n)=3n+1$), the convergent numerators satisfy*

$$p_n=(2n-1)!!\cdot(2n+1).$$

Table 2: Pi Family: even parameters (exact rationals)

c	Value	Formula
2	1	1
4	3/2	$\binom{2}{1}/2$
6	15/8	$\binom{4}{2}/2^3$
8	35/16	$\binom{6}{3}/2^4$
10	315/128	$\binom{8}{4}/2^7$
12	693/256	$\binom{10}{5}/2^8$

Proof. Base cases: $p_0 = 1 = (-1)!! \cdot 1$, $p_1 = 3 = 1!! \cdot 3$, $p_2 = 15 = 3!! \cdot 5$. Induction: assume $p_k = (2k-1)!! \cdot (2k+1)$ for $k < n$. The recurrence $p_n = (3n+1)p_{n-1} - n(2n-1)p_{n-2}$ becomes

$$(2n-1)(2n-3)P(n) = (3n+1)(2n-3)P(n-1) - n(2n-1)P(n-2)$$

where $P(n) = 2n+1$. Expanding both sides yields $0 = 0$. □ □

Theorem 6 (Conjecture 1, $m = 1$)**.** *For the Pi Family with $m = 1$ ($\alpha(n) = -n(2n-3)$, $\beta(n) = 3n+1$),*

$$p_n = (2n-1)!! \cdot (n^2 + 3n + 1).$$

Note: $n^2 + 3n + 1$ has roots $(-3 \pm \sqrt{5})/2$ related to the golden ratio.

Proof. Same structure as Theorem 5. Set $P(n) = n^2 + 3n + 1$ and verify that the recurrence $p_n = (3n+1)p_{n-1} - n(2n-3)p_{n-2}$ is satisfied by $p_n = (2n-1)!! \cdot P(n)$: dividing through by $(2n-1)!!$ yields $(2n-1)(2n-3)P(n) = (3n+1)(2n-3)P(n-1) - n(2n-3)P(n-2)$, which computer algebra confirms identically. □ □

Conjecture 7 (General m)**.** *For all $m \geq 0$ and fixed n, the quantity $R(n, m) = p_n(m)/(2n-1)!!$ is a polynomial in m of degree $\lfloor n/2 \rfloor$.*

This is verified numerically for $m = 0, \ldots, 12$ and $n \leq 20$. The detailed structure of the coefficients $c_k(n)$ in the expansion $R(n, m) = \sum_k c_k(n)\, m^k$ is discussed in Section 11.

5 The Continuous Gamma Identity

The results of Sections 2–4 are special cases of a single continuous identity. Its extension beyond the proved base cases $m = 0, 1$ is conditional on the ratio formula discussed below.

Conjecture 8 (Ratio formula)**.** *For the Wallis–PCF family, the values satisfy*

$$\frac{\mathrm{val}(m{+}1)}{\mathrm{val}(m)} = \frac{2(m{+}1)}{2m{+}1}$$

for all admissible real $m > -\frac{1}{2}$.

Theorem 9 (Conditional)**.** *The cases* $m = 0$ *and* $m = 1$ *are proved unconditionally. Assuming Conjecture 8, the PCF with* $\alpha(n) = -n(2n{-}2m{-}1)$ *and* $\beta(n) = 3n + 1$ *converges to*

$$\mathrm{val}(m) = \frac{2\,\Gamma(m{+}1)}{\sqrt{\pi}\,\Gamma(m{+}\frac{1}{2})}.$$

Proof sketch, conditional on Conjecture 8. Step 1: The raising operator. Define $L(f)_n = (n{+}2)\, f_n - (n{+}1)^2\, f_{n-1}$.

Lemma 10. *If* f *satisfies the* $m{=}0$ *recurrence* $f_n = (3n{+}1)f_{n-1} - n(2n{-}1)f_{n-2}$, *then* $g = L(f)$ *satisfies the* $m{=}1$ *recurrence* $g_n = (3n{+}1)g_{n-1} - n(2n{-}3)g_{n-2}$.

Proof of Lemma. Write $g_{n-1} = (n{+}1)f_{n-1} - n^2 f_{n-2}$ and $g_{n-2} = n\, f_{n-2} - (n{-}1)^2 f_{n-3}$. Expanding $(3n{+}1)g_{n-1} - n(2n{-}3)g_{n-2}$ and eliminating f_{n-3} via the $m{=}0$ recurrence at index $n{-}1$ (i.e. $f_{n-3} = [f_{n-1} - (3n{-}2)f_{n-2}]/[-(n{-}1)(2n{-}3)]$), one obtains coefficients

$$\begin{aligned}
&\text{coeff. of } f_{n-1} : (3n^2{+}4n{+}1) - n(n{-}1) = 2n^2{+}5n{+}1,\\
&\text{coeff. of } f_{n-2} : -n^2(5n{-}2) + n(n{-}1)(3n{-}2) = -n(2n^2{+}3n{-}2),
\end{aligned}$$

which match $g_n = (n{+}2)[(3n{+}1)f_{n-1} - n(2n{-}1)f_{n-2}] - (n{+}1)^2 f_{n-1}$ identically. □

Since L is linear, it maps the two-dimensional solution space $\mathcal{S}_0$ of the $m{=}0$ recurrence into $\mathcal{S}_1$. Checking initial values:

$$\begin{aligned}
&L(P^{(0)})_0 = 2\cdot 1 - 1\cdot 1 = 1 = P_0^{(1)}, \quad L(P^{(0)})_1 = 3\cdot 3 - 4\cdot 1 = 5 = P_1^{(1)},\\
&L(Q^{(0)})_0 = 2\cdot 1 - 1\cdot 0 = 2 = 2\,Q_0^{(1)},\ L(Q^{(0)})_1 = 3\cdot 4 - 4\cdot 1 = 8 = 2\,Q_1^{(1)}.
\end{aligned}$$

By uniqueness of solutions: $P_n^{(1)} = L(P^{(0)})_n$ and $2\,Q_n^{(1)} = L(Q^{(0)})_n$.

Step 2: Poincaré growth analysis. The recurrence $f_n = (3n{+}1)f_{n-1} - n(2n{-}2m{-}1)f_{n-2}$ has Poincaré characteristic roots $\lambda = 2$ and $\lambda = 1$ (from the leading-order equation $\lambda^2 - 3\lambda + 2 = 0$). Both $P_n^{(m)}$ and $Q_n^{(m)}$ are dominant solutions growing like $(2n)!!$, so $P_n^{(m)}/Q_n^{(m)} \to S^{(m)}$ exponentially fast. In particular, $Q_n^{(0)}/Q_{n-1}^{(0)} = (2n{+}1)(1 + O(1/n))$.

Step 3: The limit. From Step 1:

$$S_n^{(1)} = \frac{P_n^{(1)}}{Q_n^{(1)}} = 2\cdot\frac{(n{+}2)P_n^{(0)} - (n{+}1)^2 P_{n-1}^{(0)}}{(n{+}2)Q_n^{(0)} - (n{+}1)^2 Q_{n-1}^{(0)}}.$$

Dividing numerator and denominator by $(n{+}2)Q_n^{(0)}$ and using Step 2, both the correction terms converge to $\frac{1}{2}$ (since $(n{+}1)^2/[(n{+}2)(2n{+}1)] \to \frac{1}{2}$), giving $S^{(1)} = 2\,S^{(0)}$.

Step 4: General m and the Gamma identity. The base case $S^{(0)} = 2/\pi$ is established by the Gauss continued fraction for ${}_2F_1(\frac{1}{2}, 1; \frac{3}{2}; 1)$ [5]. The Wallis reduction formula gives $I_{m+1} = \frac{2m+1}{2m+2}\, I_m$ where $I_m = \int_0^{\pi/2} \sin^{2m}(x)\, dx = \frac{\sqrt{\pi}\,\Gamma(m+\frac{1}{2})}{2\,\Gamma(m+1)}$. Assuming the ratio formula $\mathrm{val}(m{+}1)/\,\mathrm{val}(m) = 2(m{+}1)/(2m{+}1) = I_m/I_{m+1}$ (verified to 200 digits for $m = 0, \ldots, 5$, and to 500 digits at $m = 0, \ldots, 10.5$), the values propagate inductively from $S^{(0)} = 1/I_0$, yielding $\mathrm{val}(m) = 1/I_m = 2\,\Gamma(m{+}1)/(\sqrt{\pi}\,\Gamma(m{+}\frac{1}{2}))$ for all non-negative integers m. Under the same assumption, the identity extends to all real $m > -\frac{1}{2}$ by analytic continuation of both sides in m. □

Corollary 11. *Assuming the ratio formula above, the following specialisations hold.*

(i) *Integer $m \geq 0$:* $\mathrm{val}(m) = 2^{2m+1}/\big(\pi\binom{2m}{m}\big)$ (Theorem 3).

(ii) *Half-integer $m = k + \frac{1}{2}$, $k \geq 0$:* $\mathrm{val}(m)$ *is an exact rational, the k-th Wallis product* (Theorem 4).

(iii) *Numerically,* $\mathrm{val}(m{+}1)/\,\mathrm{val}(m)$ *matches* $2(m{+}1)/(2m{+}1)$*, verified to* 200 *digits for* $m = 0, \ldots, 4$.

6 Meta-Family

The unified template $\alpha(n) = -(\alpha_0 n^2 + \beta_0 n)$, $\beta(n) = \gamma_0 n + \delta_0$ contains both the Logarithmic Ladder ($\beta_0 = 0$, $\delta_0 = \alpha_0$, $\gamma_0 = \alpha_0 + 1$) and the Pi Family ($\alpha_0 = 2$, $\gamma_0 = 3$, $\delta_0 = 1$) as special cases. Two new sub-families were discovered and verified to 80 digits:

- $\alpha_0 = 1$, $\beta_0 = 0$, $\gamma_0 = 4$, $\delta_0 = 2 \to 2/\ln 3$;
- $\alpha_0 = 4$, $\beta_0 = 0$, $\gamma_0 = 8$, $\delta_0 = 4 \to 4/\ln 3$.

Both are instances of the Logarithmic Ladder with non-integer k (namely $k = 3$ and the equivalence-transformed $k = 3$ at doubled scale).

7 Arithmetic Sector Bifurcation and the Congruence Hunt

The phase-1 taxonomy based on macro-spectral growth exposes a genuine organizational principle, but it is not fine enough to determine arithmetic identity. In particular, the seeds 'log-k=2' and 'pi-m=0' both lie on the

same $\Lambda = 2$ sheet. The former belongs to the logarithmic ladder, while the latter sits on the Wallis branch leading to $2/\pi$ and $4/\pi$. Thus the invariant Λ detects asymptotic scale, but not the micro-arithmetic mechanism behind the constant.

To refine the taxonomy we attach, for each prime p, the three statistics

$$D_p = \frac{1}{N}\#\{0 \le n < N : q_n \equiv 0 \pmod{p}\}, \qquad C_p = \#\{q_n \bmod p : 0 \le n < N\},$$

and

$$\omega_p = \frac{1}{\lfloor N/p \rfloor}\#\{1 \le n < N/p : q_{np} \equiv q_n \pmod{p^3}\},$$

computed here with $N = 500$ and $p \in \{2, 3, 5, 7, 11, 13\}$. The resulting signature $\Sigma = (\mathcal{D}, \mathcal{C}, \omega)$ resolves the clustering paradox.

Proposition 12 (Micro-arithmetic separation on the $\Lambda = 2$ sheet)**.** *The logarithmic seed 'log-k=2' and the Wallis seed 'pi-m=0' share the same macro-spectral value $\Lambda = 2$, but are sharply separated by the phase-2 congruence signature. At the prime $p = 11$ one finds*

$$\omega_{11}^{(\pi)} = 0.295 \qquad \textit{and} \qquad \omega_{11}^{(\log)} = 0.045,$$

while the residue-support sizes satisfy $C_{11}^{(\pi)} = 7$ and $C_{11}^{(\log)} = 10$. The sample 'vquad-3-1-1' lies far from both centroids and therefore occupies a third arithmetic sector.

Proof. This is a direct summary of the phase-2 computation recorded in Appendix A and in the generated file `results/pcf_spectral_phase2_congruence.csv`. The full vectors show that the Wallis seed has uniformly tighter residue support and stronger lifting stability than the logarithmic seed, while the V-quad sample has essentially full odd-prime support and negligible p^3-retention. □

Table 3: Selected phase-2 diagnostics for the three representative sectors.

Family	C_{11}	ω_{11}	sector reading
'log-k=2'	10	0.045	logarithmic sector
'pi-m=0'	7	0.295	Wallis/π sector
'vquad-3-1-1'	11	0.000	third-sector anomaly

The sector split is corroborated by formal verification on the Wallis side.

Theorem 13 (Formalized shift operator and Casoratian bridge)**.** *The shift operator transporting solutions of the m-level Wallis recurrence to solutions of the $(m+1)$-level recurrence, together with the discrete Casoratian*

identities used in the $\pi/4$ bridge, have been formalized in Lean 4 / Mathlib. In particular, the files `WallisFamily/ShiftConsistency.lean` *and* `WallisFamily/CasoratianPi4.lean` *machine-check the algebraic passage from Wallis-PCF convergents to the analytic Wallis product.*

Proof. This is verified by compilation of the local Lean library: the proof objects are accepted by the Lean kernel with no remaining `sorry` placeholders. □

7.1 V_{quad} as the Stokes connection constant of the cubic-coefficient ODE

The quadratic-denominator continued fraction whose Borel/PCF value is $\mathcal{V}_{\text{quad}} = 1 + \mathbf{K}_{n\geq 1} 1/(3n^2+n+1)$ is the first genuinely non-Bessel case in the hierarchy discussed above: the same recurrence-to-continuum passage that underlies the super-exponential law (1) leads to

$$(3x^2 + x + 1)\, y''(x) + (6x + 1)\, y'(x) - x^2 y(x) = 0.$$

Thus the quadratic GCF/Borel problem is accompanied by a genuine Stokes matching problem at the irregular singular point $x = \infty$, and it is natural to isolate the corresponding connection coefficient as an intrinsic analytic invariant of the cubic-coefficient ODE.

To analyze $x = \infty$, insert the WKB ansatz $y(x) \sim x^{\mu} e^{\sigma x}$ into the differential equation. The x^2-balance gives $3\sigma^2 - 1 = 0$, hence $\sigma_{\pm} = \pm 1/\sqrt{3}$; matching the next x-order terms yields $6\sigma\mu + \sigma^2 + 6\sigma = 0$, so

$$\mu_+ = -1 - \frac{1}{6\sqrt{3}}, \qquad \mu_- = -1 + \frac{1}{6\sqrt{3}}.$$

We therefore define the formal dominant branch (which grows like $e^{x/\sqrt{3}}$ as $x \to +\infty$) and the recessive branch (which decays like $e^{-x/\sqrt{3}}$) on the positive real axis by

$$y_{\text{dom}}(x) \sim x^{\mu_+} e^{x/\sqrt{3}} \sum_{n\geq 0} a_n x^{-n}, \qquad y_{\text{rec}}(x) \sim x^{\mu_-} e^{-x/\sqrt{3}} \sum_{n\geq 0} b_n x^{-n},$$

with $a_0 = b_0 = 1$.

At $x = 0$ the indicial roots are 0 and 1, so there are no logarithmic terms and one has an analytic Frobenius basis $y_1(x) = 1 + O(x^4)$, $y_2(x) = x + O(x^2)$ with $y_1(0) = 1$. We define the Stokes connection constant V_{quad} by the coefficient of y_1 in the exact local expansion

$$y_{\text{rec}}(x) = V_{\text{quad}}\, y_1(x) + C_{\text{quad}}\, y_2(x)$$

near $x = 0$; equivalently, if

$$\begin{pmatrix} y_{\text{rec}} & y_{\text{dom}} \\ y'_{\text{rec}} & y'_{\text{dom}} \end{pmatrix} = \begin{pmatrix} y_1 & y_2 \\ y'_1 & y'_2 \end{pmatrix} M$$

at the matching point $x_{\mathrm{match}} = 10$, then $V_{\mathrm{quad}} = M_{11}$. Numerically,

$$V_{\mathrm{quad}} = M_{11} = 1.9420321374711220465\ldots,$$

while the second Frobenius coefficient satisfies

$$M_{21} = -2.9999268666050110215\ldots = -3 + \delta, \qquad \delta \approx 7.313 \times 10^{-5},$$

and the 500-digit rerun confirms that this small deviation from -3 is genuine rather than numerical noise. This is distinct from the raw transported sectorial value $y_{\mathrm{rec}}(0) = 1.94257302477940314397\ldots$ (with $y'_{\mathrm{rec}}(0)/y_{\mathrm{rec}}(0) = -1.54473580935967544255\ldots$) obtained from the earlier RK4 Stokes integration at dps= 300: that quantity depends on the normalization of y_{rec}, whereas $V_{\mathrm{quad}} = M_{11}$ is the normalization-independent coefficient of y_1.

Theorem 14. *The Stokes connection constant V_{quad} does not satisfy any non-trivial integer-linear relation with coefficients bounded by* 500 *in the explicit basis*

$$\begin{aligned}\mathcal{B} = \{&1,\ \pi,\ \log 2,\ \log 3,\ \sqrt{3},\ \Gamma(1/3),\ \Gamma(2/3),\ \Gamma(1/4),\ \Gamma(3/4),\ \zeta(3),\\ &{}_0F_2(;2/3,4/3;4/27),\ {}_0F_2(;1/3,2/3;4/27),\ {}_0F_2(;1/3,4/3;4/27),\\ &{}_0F_2(;2/3,5/3;4/27),\ {}_0F_2(;1/3,2/3;1/27),\ {}_0F_2(;2/3,4/3;1/27),\\ &\mathrm{Ai}\Big((11/12)^{1/3}\Big),\ \mathrm{Bi}\Big((11/12)^{1/3}\Big),\ \mathrm{Ai}\Big(-(11/12)^{1/3}\Big),\ \mathrm{Bi}\Big(-(11/12)^{1/3}\Big),\\ &\mathrm{Ai}(3^{-1/3}),\ \mathrm{Bi}(3^{-1/3}),\ \mathrm{Ai}(11^{1/3}/3),\ \mathrm{Bi}(11^{1/3}/3),\\ &\mathrm{Ai}\Big((\sqrt{11}/6)^{2/3}\Big),\ \mathrm{Bi}\Big((\sqrt{11}/6)^{2/3}\Big),\\ &I_{1/3}\Big(\tfrac{2}{3\sqrt{3}}\Big),\ K_{1/3}\Big(\tfrac{2}{3\sqrt{3}}\Big),\ I_{2/3}\Big(\tfrac{2}{3\sqrt{3}}\Big),\ K_{2/3}\Big(\tfrac{2}{3\sqrt{3}}\Big),\\ &I_{1/3}\big(\tfrac{2}{3}\big),\ K_{1/3}\big(\tfrac{2}{3}\big)\}.\end{aligned}$$

This was verified via the PSLQ algorithm with `maxcoeff`*=500 at* `dps`*=300.*

Structurally, this is the same phenomenon seen for Stokes multipliers in Painlevé II/III and in confluent-Heun problems, where the connection constants are typically more primitive than isolated classical special values. In that sense V_{quad} behaves less like a disguised Gamma-ratio and more like an entry of a global monodromy/Stokes matrix. The obstruction to a direct ${}_0F_2$ reduction is already visible in the Maclaurin recurrence, whose c_{n-2} term prevents the first-order Pochhammer structure of a single generalized hypergeometric solution. The persistent PSLQ failures therefore point to structure, not to numerical accident. The natural next step is a full symbolic monodromy computation or a connection to the theory of resurgence/alien calculus.

8 Quadratic GCF Constants

The phase-2 congruence hunt shows that the V-quad constants are not merely unidentified values within the old taxonomy: they form a structured third sector, or arithmetic *archipelago*, invisible to the macro invariant Λ alone. Their residue-support profile is markedly flatter than that of the Wallis/π branch, and their lifting stability is essentially absent at the odd primes; see Appendix A.

Define $V(A,B,C) = 1 + \mathbf{K}_{n\geq 1}\, 1/(An^2+Bn+C)$. The original $V_{\text{quad}} = V(3,1,1)$ has discriminant -11.

Proposition 15. *For all (A,B,C) with $A \geq 1$, $C \geq 1$, and $An^2+Bn+C > 0$ for $n \geq 1$, $V(A,B,C)$ is irrational.*

Proof. The convergent denominators Q_n satisfy $Q_n = (An^2+Bn+C)Q_{n-1} + Q_{n-2}$ with $Q_0 = 1$, $Q_{-1} = 0$. Since $An^2+Bn+C \geq 1$ for all $n \geq 1$, Q_n grows at least exponentially: $\log Q_n \gtrsim n \log n$. The Wronskian $|P_n Q_{n-1} - P_{n-1} Q_n| = 1$ for all n. By Euler's irrationality criterion, the limit is irrational. □ □

Our systematic scan found **482 distinct** such constants. A PSLQ integer relation search at 300-digit precision over an extended basis of 15 constants $\{\pi, \pi^2, e, \ln 2, \ln 3, \gamma, G, \zeta(3), \zeta(5), \sqrt{2}, \sqrt{3}, \sqrt{5}, \varphi, \sqrt{\pi}, \pi \ln 2\}$ finds no relation for any representative V-constant, and none is algebraic of degree ≤ 8 with coefficients $\leq 10^5$. A targeted sweep against 588 expressions built from Gamma ratios, Pi Family values $\mathrm{val}(m)$, and standard-constant multiples also produces no matches; in particular, no V-constant equals $\mathrm{val}(m)$ for any real m. Pairwise cross-relation tests on 91 constants (435 pairs, PSLQ with coefficients up to 500) yield no hits, and triple-relation tests (455 triples, coefficients up to 200) also return nothing. Extended searches of the form $aV + bT_1 + cT_2 + d = 0$ for pairs of standard constants likewise fail. The only structural relation found is the *shift symmetry* $V(1,B,C) - V(1,-B,C) \in \mathbb{Z}$ for all tested B, C, which follows from the palindromic structure of the GCF denominators. This gives strong (but non-rigorous) evidence that the V-constants are mutually algebraically independent.

Remark 16. An earlier PSLQ search returned the relation $[0,1,-2,1]$ relative to $\{V, \sqrt{5}, \varphi, 1\}$; however, the coefficient of V is zero—this is the trivial identity $\sqrt{5} - 2\varphi + 1 = 0$ and does not involve V.

8.1 Discriminant -11 and a confluent-Heun obstruction

For the equivalence-transformed recurrence with $a_n = -n^2$ and $b_n = 3n^2 + n + 1$, the Birkhoff–Trjitzinsky continuum limit gives the second-order differential equation

$$(3x^2 + x + 1)\, y''(x) + (6x+1)\, y'(x) - x^2 y(x) = 0.$$

Its finite singularities are exactly the CM points

$$r_\pm = \frac{-1 \pm i\sqrt{11}}{6},$$

while $x = 0$ is an ordinary point and $x = \infty$ is irregular. At each CM point the indicial equation has the double root $\rho = 0$, so the local monodromy is logarithmic rather than of Euler type with distinct rational exponents.

Under the Möbius normalization

$$z = \frac{x - r_+}{r_- - r_+},$$

which sends $r_+ \mapsto 0$, $r_- \mapsto 1$, and $x = 0$ to $z_0 = \frac{1}{2} + \frac{i}{2\sqrt{11}}$, the equation becomes

$$y_{zz} + \left(\frac{1}{z} + \frac{1}{z-1}\right) y_z + \left(\frac{11}{27} + \frac{A}{z} + \frac{B}{z-1}\right) y = 0,$$

with

$$A = \frac{-5 - i\sqrt{11}}{54}, \qquad B = \frac{5 - i\sqrt{11}}{54}.$$

After the gauge transformation $y = e^{\lambda z} u$ with $\lambda^2 = -11/27$, one obtains

$$u'' + \left(\alpha + \frac{1}{z} + \frac{1}{z-1}\right) u' + \left(\frac{\mu}{z} + \frac{\nu}{z-1}\right) u = 0,$$

which is of confluent-Heun type rather than the ordinary Heun equation with four regular singular points. Thus the discriminant -11 does encode genuine CM geometry, but the associated special-function object is a connection coefficient problem for a confluent equation, not a direct $\mathrm{HeunG}(a, q, \alpha, \beta, \gamma, \delta; z_0)$ evaluation.

A 300-digit PSLQ test against the CM basis $\{1, \pi/\sqrt{11}, \pi^2/11, \Gamma(1/11)\Gamma(2/11)\Gamma(4/11)/\Gamma(8/11), 2\pi/\sqrt{11}, \sqrt{11}, 11^{1/4}, \log|\eta((1 + i\sqrt{11})/2)|\}$ finds no non-trivial relation involving V_{quad} [computed, `mpmath`, dps= 300].

8.2 ${}_0F_2$, Airy, Bessel, and Stokes diagnostics

At the irregular point $x = \infty$, the WKB ansatz $y(x) \sim x^\mu e^{\sigma x}$ gives the exact exponents

$$\sigma_\pm = \pm\frac{1}{\sqrt{3}}, \qquad \mu_\pm = -1 \mp \frac{1}{6\sqrt{3}}, \tag{1}$$

so the dominant/recessive branches are $y_\pm(x) \sim x^{\mu_\pm} e^{\pm x/\sqrt{3}}$. If $y(x) = \sum_{n\geq 0} c_n x^n$, then the differential equation yields the coefficient recurrence

$$(n+2)(n+1)c_{n+2} + (n+1)^2 c_{n+1} + 3n(n+1)c_n - c_{n-2} = 0,$$

which is not the first-order Pochhammer recurrence of a single ${}_0F_2$ series. Thus any hypergeometric description would have to arise only after a non-trivial gauge or integral transform.

We tested the six natural values $F_j \in \{{}_0F_2(;2/3,4/3;4/27), {}_0F_2(;1/3,2/3;4/27), {}_0F_2(;1/3,4/3;4/2$ ${}_0F_2(;2/3,5/3;4/27), {}_0F_2(;1/3,2/3;1/27), {}_0F_2(;2/3,4/3;1/27)\}$, all Airy values $\mathrm{Ai}(z), \mathrm{Bi}(z)$ at the CM-motivated arguments listed in the computational appendix, and the Bessel values $I_{1/3}, K_{1/3}, I_{2/3}, K_{2/3}$ at $2/(3\sqrt{3})$ and $2/3$. At 300-digit precision, PSLQ finds no non-trivial relation involving V_{quad} in any of these bases [computed, `mpmath`, dps= 300; maxcoeff 500 for the ${}_0F_2$ sweep and 200 for the Airy/Bessel sweeps]. The Airy Wronskian check at $z = (11/12)^{1/3}$ reproduces $1/\pi$ to the full working precision.

Finally, integrating the recessive WKB branch from $x_0 = 1000$ back to $x = 0$ with initial data $y(x_0) = x_0^{\mu_-} e^{-x_0/\sqrt{3}}$ gives the numerically stable connection values

$$y(0) = 1.9425730247794031439\ldots, \qquad \frac{y'(0)}{y(0)} = -1.5447358093596754425\ldots,$$

and hence

$$\frac{y(0)}{y'(0)} = -0.6473598876525821160\ldots.$$

These do not agree with V_{quad} or $1/V_{\mathrm{quad}}$ up to any small rational factor detected by PSLQ, so the Stokes multiplier remains an apparently new constant rather than a low-height special-function value.

8.3 A hardened three-basis confluent-Heun/Stokes exclusion

We next tested whether V_{quad} becomes visible only after coupling the confluent-Heun sector to the Stokes data and the discriminant-11 CM/η data. In the notation of Section 8.1, the normalized equation is

$$u'' + \left(\alpha + \frac{1}{z} + \frac{1}{z-1}\right) u' + \left(\frac{\mu}{z} + \frac{\nu}{z-1}\right) u = 0,$$

with

$$z_0 = \texttt{0.5 + 0.150755672288881811323406 i}, \qquad \alpha = \frac{2i\sqrt{11}}{3\sqrt{3}},$$

$$\mu = \frac{-5 - i\sqrt{11}}{54} + \frac{i\sqrt{11}}{3\sqrt{3}}, \qquad \nu = \frac{5 - i\sqrt{11}}{54} + \frac{i\sqrt{11}}{3\sqrt{3}},$$

where the CM motivation is the discriminant-11 η-geometry at $\tau = i\sqrt{11}/3$.

The numerical bottleneck in this search is the evaluation of the regular local Heun branches at z_0. An earlier RK4 path transport gave only `7.642` digits of branch stability, so that computation was useful only as a heuristic

probe. We therefore replaced it by local regular-series recurrences about $z = 0$ and $z = 1$, matched at z_0; at `120` decimal digits and continued-fraction depth `9000`, the two independent local evaluations now agree to `35.207` digits. Since this stability margin exceeds the best observed near-miss by almost 30 digits, the null PSLQ output is no longer plausibly attributable to solver drift.

We then searched for low-height relations

$$aV_{\text{quad}} + bH + cS + dC + e = 0,$$

where H ranges over the regular confluent-Heun invariants at z_0, S over $\{y(0), y'(0), y'(0)/y(0), 1/(y'(0)/y(0)), M_{11}, M_{21}, M_{11}/M_{21}, M_{12}/M_{22}\}$, and C over the CM basis $\{\pi/\sqrt{11}, \pi^2/11, \Gamma(1/11)\Gamma(2/11)\Gamma(4/11)/\Gamma(8/11), \sqrt{11}, \log|\eta(\tau)|\}$. A total of `360` triples were tested with PSLQ coefficient bound `2000`; the number of non-trivial hits involving V_{quad} was `0`. The strongest low-height near-miss was

$$V_{\text{quad}} \approx -\Im\left(\frac{H_0'(z_0)}{H_0(z_0)}\right) + \frac{3}{2}\,y(0) + 3\log|\eta(\tau)| - 1, \tag{2}$$

with accuracy only `5.507602` digits.

For the exclusion table: the hardened three-basis confluent-Heun/Stokes/η search at `120` dps gives `360` tested triples, `0` non-trivial PSLQ hits, and best near-miss `5.507602` digits.

8.4 A bridged fixed-alpha family with affine skeleton

We now isolate the most rigid one-parameter family that emerged from the quadratic search. Its leading term is completely explicit, and the remaining nontrivial structure is pushed into a numerically small tail.

Definition 17 (Bridged fixed-alpha family)**.** For $m \geq 0$, let

$$x(m) = b_0(m) + \mathbf{K}_{n=1}^{\infty} \frac{a_n(m)}{b_n(m)}$$

be the generalized continued fraction attached to the label `fixed_alpha` with parameters $(a_1, a_2, b_1, b_2, C, \text{bridge}) = (-15, -1, -4, 0, 5/4, +1)$, namely

$$a_n(m) = -\tfrac{5}{4}\,n^6, \tag{3}$$

and

$$b_n(m) = (2n+1)\big((17-15m-m^2)n^2+(17-15m-m^2)n+(5-4m)\big)+(3n^2+n+1), \tag{4}$$

with constant term

$$b_0(m) = (5 - 4m) + 1 = 6 - 4m. \tag{5}$$

Proposition 18 (Affine skeleton)**.** *For every $m \geq 0$ for which the continued fraction in Theorem 17 converges, its value admits the exact decomposition*

$$x(m) = 6 - 4m + \varepsilon(m), \tag{6}$$

where the correction term is precisely the continued-fraction tail

$$\varepsilon(m) = \mathbf{K}_{n=1}^{\infty} \frac{a_n(m)}{b_n(m)}. \tag{7}$$

Proof. This is immediate from the definition of a generalized continued fraction. The head term is exactly $b_0(m) = (5-4m)+1 = 6-4m$, so subtracting the head from $x(m)$ leaves the tail (7). No approximation is involved. □

Remark 19. Forward/backward Pincherle consistency for the representative instances used in this family was verified numerically to about 120 digits [computed, `mpmath`, dps= 120]. For $m \geq 2$ and all tested $m \leq 50$, the correction term in Equation 6 is positive and satisfies $0 < \varepsilon(m) < 0.03$; e.g. $\varepsilon(2) = 0.0118099517\ldots$ and $\varepsilon(50) = 0.0000625688\ldots$ [computed, `mpmath`, dps= 400]. The exact asymptotic form of $\varepsilon(m)$ remains open. Current data suggest a power-law decay with an exponent near 3/2, but the local exponent continues to drift as m grows, consistent with a possible logarithmic correction of the form $\varepsilon(m) \sim C\, m^{-\alpha}(\log m)^{\beta}$. In particular, a PSLQ exclusion at 350 digits with coefficient bound 5000 finds no relation for the fitted coefficient C in bases generated by $\{\Gamma(1/4), \zeta(3), \pi, \sqrt{2}, \log(1+\sqrt{2})\}$ and low-degree products thereof [computed, `mpmath`, dps= 350].

8.5 The Apéry Basin

A companion specialization of the same search architecture recovers the classical Apéry kernel. Here the parameters are $(a_1, a_2, b_1, b_2, C, \text{bridge}) = (-15, 0, -4, 0, 1, 0)$, so that

$$a_n = -n^6, \qquad b_n = (2n+1)(17n^2 + 17n + 5). \tag{8}$$

This is exactly the standard cubic kernel from Apéry's proof of the irrationality of $\zeta(3)$ [2].

At $m = 0$ we obtain the numerically rigid identification

$$x(0) = \frac{6}{\zeta(3)} = 4.9914442355\ldots$$

with Pincherle agreement to 119.5 digits and Bayesian evidence ratio approximately 8.64×10^5 [computed, `mpmath`, dps= 500]. A direct PSLQ computation on the basis $\{x, 1/\zeta(3), 1\}$ returns the relation vector $[-1, 6, 0]$ with residual 0.0 [computed, `mpmath`, dps= 500]. Thus the basin value is firmly identified as $6/\zeta(3)$, even though the convergents are not termwise identical to the classical Apéry sequences A_n/B_n; rather, they realize the same kernel on the reciprocal/Pincherle side.

Remark 20. A closed-form derivation via the classical hypergeometric ${}_3F_2(1)$ mechanism is strongly suggested by the data, but is not yet proved in the present paper. We therefore record the $6/\zeta(3)$ evaluation as a computationally certified companion result rather than a completed analytic identity.

9 A Cubic Clausen–Trigonometric Witness

A particularly rigid experimental witness arises from the cubic generalized continued fraction

$$\begin{aligned} B_{\text{cubic}} &= \text{eval_cf}(2,3,2,2/3) \\ &= \frac{2}{3} + \mathbf{K}_{n=1}^{\infty} \frac{2n^3}{2n+\frac{2}{3}} = 1.11003023075477593869\ldots. \end{aligned}$$

Here the continued-fraction convention is

$$b_0 + \cfrac{a_1}{b_1 + \cfrac{a_2}{b_2 + \cdots}}, \qquad a_n = 2n^3, \quad b_n = 2n + \frac{2}{3}, \quad b_0 = \frac{2}{3}.$$

This value is the $\beta = 2/3$ branch of the cubic ridge discovered in the broad GDCF scan. After negative tests against standard transcendental bases, conductor-36 CM periods, genus-2 periods, Mahler measures, Apéry-like sums, and rational ${}_3F_2$ grids, a continuous hypergeometric search produced a one-parameter fit of remarkable rigidity.

Conjecture 21 (Cubic Clausen-square identification)**.** *There exists a unique real parameter $u \in (0, \frac{1}{2})$, numerically*

$$u = 0.18852361548552202250386494023318724958999804944\ldots,$$

such that

$$B_{\text{cubic}} = {}_2F_1\big(u, 1-u; \tfrac{3}{2}; \tfrac{27}{64}\big)^2.$$

Equivalently, if $\lambda = 1 - 2u$ and $\theta = \arcsin(3\sqrt{3}/8)$, then

$$B_{\text{cubic}} = \left(\frac{\sin(\lambda\theta)}{\lambda \sin\theta}\right)^2.$$

A companion evaluation at $\beta = \frac{1}{3}$ gives

$$B\big(\tfrac{1}{3}\big) = 0.805664740414447796515\ldots = {}_2F_1\big(u', 1-u'; \tfrac{3}{2}; \tfrac{27}{64}\big)^2,$$

with $u' = -0.24589471253854869965\ldots$ on the $\lambda > 1$ sheet of the same cubic modular point $z = 27/64$.

Remark 22. Exhaustive integer-relation searches at 300-digit precision, including `findpoly` to degree 20, PSLQ over mixed bases involving $\sqrt{3}$ and the elliptic periods $K(k), K(k')$ with $k = 3\sqrt{3}/8$, and pairwise combination tests over all five computed λ-values, find no algebraic structure for the deformation parameter $\lambda(\beta)$.

Remark 23 (Structural map of the point 27/64). The current evidence suggests that $z = 27/64$ is not selected by a classical singularity of the recurrence itself. For the convergents $X_n \in \{P_n, Q_n\}$,

$$X_n = (2n + \beta)X_{n-1} + 2n^3 X_{n-2},$$

so the local characteristic roots are

$$r_\pm(n;\beta) = \frac{2n + \beta \pm \sqrt{(2n+\beta)^2 + 8n^3}}{2} = \pm\sqrt{2}\, n^{3/2} + n + O(\sqrt{n}),$$

and therefore

$$\left|\frac{r_+(n;\beta)}{r_-(n;\beta)}\right| = 1 + O(n^{-1/2})$$

for every real β. Thus the cubic family is already asymptotically on a neutral/Stokes boundary, and the "spectral gap closing" is a feature of the whole ridge rather than something unique to $\beta = 2/3$.

Instead, the numerics point to a different bottleneck: one can keep the hypergeometric argument fixed at $z = 27/64$ and vary only the monodromy parameter $\lambda(\beta)$ in

$$\text{eval_cf}(2,3,2,\beta) = \left(\frac{\sin(\lambda(\beta)\,\theta)}{\lambda(\beta)\sin\theta}\right)^2, \qquad \theta = \arcsin(3\sqrt{3}/8),$$

with $\lambda(1/3) \approx 1.4917894$, $\lambda(1/2) \approx 1.1185647$, and $\lambda(2/3) \approx 0.62295277$.

A concrete cubic explanation is encoded by the Chebyshev-type resolvent

$$4x^3 - 3x + \frac{8}{3\sqrt{3}}\sqrt{z} = 0,$$

whose discriminant with respect to x is

$$\Delta_x = -16(64z - 27) = 432\left(1 - \frac{64z}{27}\right).$$

Hence $z = 27/64$ is exactly the ramification point of the cubic pullback. Equivalently, with

$$t = \sqrt{1 - \frac{64z}{27}}, \qquad z = \frac{27}{64}(1 - t^2),$$

the pulled-back Gauss equation for ${}_2F_1(u, 1-u; \frac{3}{2}; z)$ acquires a regular singularity at $t = 0$. This also explains the integers 27 and 64: 27 is the signature-3 / triplication scale coming from the cubic combinatorics

$$(\tfrac{1}{3})_n(\tfrac{2}{3})_n = \frac{(3n)!}{27^n n!},$$

while $64 = 4^3$ is the matching Chebyshev normalization in the cubic discriminant.

In this variable the real phase portrait is

$$0 \le z < \frac{27}{64} \iff t \in (0, 1], \qquad z = \frac{27}{64} \iff t = 0, \qquad z > \frac{27}{64} \iff t \in i\mathbb{R}.$$

So the subcritical regime supports the real trigonometric branch, the critical value $z = 27/64$ is the Goldilocks ramification point where the cubic sheets merge, and the supercritical regime forces hyperbolic/complex continuation rather than a new real Clausen square. The real branch persists up to a numerically observed endpoint

$$\beta_{\text{crit}} \approx 0.7478702596, \qquad \lambda(\beta_{\text{crit}}) = 0,$$

where the value saturates at $(\theta/\sin\theta)^2 \approx 1.18466547096$. So $\beta = 2/3$ is an interior point of a smooth phase line, not the phase transition itself. Likewise, denominator-growth / error-ratio proxies vary smoothly with β and show no isolated Apéry-style spike at $\beta = 2/3$.

Finally, the associated Legendre invariant

$$j(27/64) = \frac{256(1 - z + z^2)^3}{z^2(1-z)^2}\Big|_{z=27/64} = \frac{29704593673}{15968016} \approx 1860.2557558$$

is rational but not an algebraic integer, so $z = 27/64$ is not a classical CM singular modulus. In this sense the point behaves more like a *modular shadow*: a rigid cubic pullback anchor for the hypergeometric representation, rather than a genuine cusp or singular point of the continued fraction.

10 Toward Signature-4 Clausen–Square Collapses: The Quarter-Shift Family

The cubic pair from Section 9 suggests looking not only at higher degree in the denominator polynomial but also at quarter-shifts of the same $\text{eval_cf}(2, 3, 2, \beta)$ ridge. The strongest new signature-4 bridge is the lower-quarter branch

$$\begin{aligned} Q_{1/4} &= \text{eval_cf}(2, 3, 2, 1/4) \\ &= \frac{1}{4} + \mathbf{K}_{n=1}^{\infty} \frac{2n^3}{2n + \frac{1}{4}} = 0.730120743284627545239474403807\ldots. \end{aligned}$$

Conjecture 24 (Quartic contiguous-square bridge)**.** *There exists a real parameter $u_Q < 0$, numerically*

$$u_Q = -0.669740668678523125004130174543817230105714497894\ldots,$$

such that

$$Q_{1/4} = {}_2F_1\big(u_Q, 1-u_Q; 2; \tfrac{1}{4}\big)^2.$$

Fresh high-precision verification gives residual below 10^{-271} and depth stability better than 10^{-211} between truncation levels 30000 and 50000.

Numerically this is not an isolated point. Solving

$$\mathrm{eval_cf}(2,3,2,\beta) = {}_2F_1\big(u(\beta), 1-u(\beta); 2; \tfrac{1}{4}\big)^2$$

for sampled $\beta \in [1/4, 1/3]$ yields a smooth monotone branch

$$u(1/4) \approx -0.6697407, \quad u(4/15) \approx -0.6390304, \quad u(17/60) \approx -0.6079796, \quad u(3/10) \approx -0.5765408,$$

$$u(19/60) \approx -0.5446612, \quad u(1/3) \approx -0.5122818.$$

This provides a concrete computational bridge from the quartic point $z = 1/4$ to the companion cubic branch.

The upper-quarter branch does not fit the bare square, but it does admit a natural prefactored continuation. For

$$Q_{3/4} = \mathrm{eval_cf}(2,3,2,3/4) = 1.18662534630505664722042010001\ldots,$$

one finds

$$Q_{3/4} = \frac{4}{3}\,{}_2F_1\big(\widetilde{u}, 1-\widetilde{u}; 2; \tfrac{1}{4}\big)^2, \qquad \widetilde{u} \approx -0.319418247512852660858342950669600 06\ldots,$$

again with residual below 10^{-180} in fresh 180-digit checks. Thus the quarter-shift family appears to split into an unprefactored lower branch and a rationally prefactored upper branch, both tied to the same quartic modular point $z = 1/4$.

11 Further Structure of the Convergent Numerators

Define $R(n,m) = p_n(m)/(2n-1)!!$ as in Conjecture 1. For $m = 0$ the closed form is $R(n,0) = 2n+1$, equivalently $2\,\Gamma(n+\frac{3}{2})/\Gamma(n+\frac{1}{2})$—elementary, but the continuous extension to all real m (Theorem 9) is not. For $m = 1$, $R(n,1) = n^2+3n+1$ (OEIS A028387 [9], discriminant 5), whose reciprocal sum satisfies the identity

$$\sum_{n=1}^{\infty} \frac{1}{n^2+3n+1} = 1 + \frac{\pi \tan(\sqrt{5}\,\pi/2)}{\sqrt{5}}.$$

For $m \geq 2$, $R(n,m)$ is rational with denominators dividing the product of odd primes $\leq 2n-1$. As a function of m for fixed n, $R(n,m)$ is polynomial of degree $\lfloor n/2 \rfloor$ with coefficients $c_k(n)$ satisfying $c_0(n) = 2n+1$ universally and, for $k \geq 1$,

$$\operatorname{den}(c_k(n)) \,\Big|\, \prod_{\substack{p \leq 2n-1 \\ p \text{ odd prime}}} p^{a_p(n)}.$$

This structure is consistent with a hypergeometric representation of $R(n,m)$ involving half-integer parameters, but the exact closed form remains open.

12 Arithmetic Sector Separation and Methodological Outlook

The failure of universal convergence

One of the clearest messages of the present study is that convergence in $\mathbb{R}$ does not imply membership in a single arithmetic class. Every PCF considered here converges in the Archimedean metric, and several do so with exceptional numerical stability. The extended local scan nevertheless detects a persistent *congruence obstruction*—equivalently, a residual lifting deficiency—for the anomalous V-Quad sector at the primes

$$p \in \{17, 19, 23\}.$$

This defect is absent from the proved Wallis/π and logarithmic tracks. The resulting picture is summarized by Figure 1: the verified families form a tight arithmetic "mainland," while the V-Quad points occupy a detached cluster. In this sense the families belong to disjoint arithmetic universes even when their real limits display comparable stability.

The contrast is sharpened by the formal side of the project. The shift-operator mechanism underlying Theorem 9, together with the $m = 1$ Casoratian-to-series bridge, now admits a Lean 4 formalization without sorry placeholders; the experimental identities on the Wallis track therefore pass from conjectural fits to theorem-grade objects. The V-Quad sector behaves differently. Its real convergence remains robust, but the local obstruction at $p = 17, 19, 23$ prevents it from fitting naturally into the same arithmetic regime.

The V-Quad archipelago

We therefore propose to view the anomalous V-Quad constants as a provisional *V-Quad Archipelago*: not a single outlier, but a cluster of numerically coherent values that remains detached from the Wallis/logarithmic mainland. At present these constants evade both the usual logarithmic templates and the Casoratian/π-type mechanisms proved above. One plausible

interpretation is that they behave as *non-periods*, or as quantities influenced by exponential-motivic phenomena[1], which would explain their rapid real convergence alongside poor p-adic purity. This classification remains provisional, but the detached clustering and congruence obstruction already justify treating the V-Quad sector as a genuinely new arithmetic zone rather than as a deformed member of the known families.

Automated conjecture-to-theorem pipelines

Methodologically, the paper also records a shift in practice. The SIARC v6 agent-discovery stack can scan large PCF parameter spaces, isolate rigid candidates, and rank them by numerical purity and stability; Lean 4 and `mathlib` [3, 4] can then certify the symbolic mechanisms that survive this sieve. The result is not a replacement for proof, but a disciplined pipeline from experimental signal to theorem-grade certainty. The Wallis track confirms that automated exploration can culminate in machine- checked mathematics, while the V-Quad archipelago shows that the same machinery also serves as a compass at the frontier, detecting where classical classification ends and new arithmetic begins.

13 Open Questions

The following questions arise directly from the numerical evidence and remain open.

1. **Proof of the Clausen-square identity.** Prove that $B(2/3) = [{}_2F_1(u, 1-u; \frac{3}{2}; \frac{27}{64})]^2$ for $u = 0.18852361548552202250\ldots$ as an exact identity, not merely a high-precision numerical coincidence. The classical Clausen identity provides a structural scaffold; what is missing is a direct derivation connecting $\mathrm{eval_cf}(2, 3, 2, \beta)$ to the hypergeometric value at this argument.

2. **Nature of the deformation parameter $\lambda(\beta)$.** For each β in the real branch $[0, \beta_{\mathrm{crit}}]$, the map $\beta \mapsto \lambda(\beta)$ is smooth and monotone decreasing. Integer-relation searches at 300-digit precision find no algebraic relation for $\lambda(\beta)$ at degree ≤ 20 with coefficient height $\leq 10^5$, no relation involving $\sqrt{3}$ or the elliptic periods $K(k), K(k')$ for $k = 3\sqrt{3}/8$, and no relation of the form $\lambda(\beta)\theta/\pi \in \mathbb{Q}$. Whether $\lambda(\beta)$ is transcendental for all rational β, or algebraic at a sparse set of special values, remains open.

[1] This is meant heuristically rather than as a formal classification claim: the point is that the V-Quad sector appears to fall outside the classical period templates that govern the proved Wallis and logarithmic families.

3. **Non-classical modular status.** The elliptic modulus $k = 3\sqrt{3}/8$ satisfies $k^3 + (k')^3 = 0.71359\ldots \neq 1$, placing $z = 27/64$ strictly outside the Borwein signature-3 singular locus. The associated nome $q = 0.034194\ldots$ matches no entry $q_r = e^{-\pi\sqrt{r}}$ for $r \leq 36$. What modular or arithmetic object, if any, organizes the selection of $z = 27/64$ by the continued-fraction family $\mathrm{eval_cf}(2, 3, 2, \beta)$?

4. **The real-branch endpoint.** The map $\beta \mapsto V(\beta)$ appears to have a critical point near $\beta_{\mathrm{crit}} \approx 0.74787025960$, where $V(\beta_{\mathrm{crit}}) \approx (\theta/\sin\theta)^2$, the $\lambda \to 0$ limit of the trigonometric form. Does this endpoint admit a closed form, and does the family admit a natural complex or hyperbolic continuation beyond it?

5. **Higher-degree analogues.** Does an analogous Clausen-square collapse occur for $\mathrm{eval_cf}(p, d, \alpha, \beta)$ with $d \geq 4$? A preliminary scan at the quartic point $z = 1/4$ for $\mathrm{eval_cf}(1, 4, 7, \beta)$ shows a weak match at five-digit precision; this deserves systematic high-precision investigation.

A Phase-2 Congruence Signature Table

Table 4 records the raw congruence signatures for the three representative seeds highlighted in Section 7. These values were generated automatically by the phase-2 residue scan and are the data-level explanation for the arithmetic sector split.

Table 4: Raw phase-2 congruence signatures for the seed sectors (500 terms; primes $2, 3, 5, 7, 11, 13$).

Family	$\mathcal{D}$ (zero-density vector)	$\mathcal{C}$ (support sizes)	ω (mod p^3 retention)
'log-k=2'	$[0.996, 0.990, 0.984, 0.974, 0.958, 0.903]$	$[3, 5, 6, 10, 11]$	$[0.984, 0.939, 0.838, 0.629, 0.045, 0.000]$
'pi-m=0'	$[0.998, 0.992, 0.988, 0.980, 0.970, 0.962]$	$[2, 4, 6, 7, 11]$	$[0.992, 0.952, 0.848, 0.671, 0.295, 0.000]$
'vquad-3-1-1'	$[0.332, 0.332, 0.266, 0.124, 0.082, 0.076]$	$[3, 5, 7, 11, 13]$	$[0.040, 0.073, 0.030, 0.000, 0.000, 0.000]$

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Results discovered autonomously through the Multi-Agent Conjecture Discovery Framework (Rounds 2–10). All computations are reproducible with the attached Python notebooks. AI assistance disclosure. The preparation of this manuscript involved assistance from Claude (Anthropic, version claude-sonnet-4-6) for LaTeX editing, proof review, and computational interpretation. All mathematical results, proofs, and numerical verifications were conceived, directed, and validated by the author. The AI was used as a writing and research tool, not as a source of original mathematics.

Declaration of Generative AI Use

During the preparation of this manuscript the author used GitHub Copilot (Microsoft, GPT-4-based) and Claude (Anthropic, claude-sonnet-4-6) to assist with the following tasks: reorganisation and consolidation of LaTeX source files; drafting and editing of the abstract and introduction; cross-reference auditing and repair; and synthesis of connections across the five constituent manuscripts. All mathematical content, proofs, computational results, and scientific conclusions were generated, verified, and remain the sole responsibility of the author. No AI tool was used to generate or manipulate figures, tables, or research data.

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